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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

BT1302-1 – BIOMEDICAL INSTRUMENTATION
(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Power in an electrical circuit is expressed in:
A) Joules
B) Watts
C) Amperes
D) Volts
2. Which of the following is NOT a functional element of an instrumentation system?
A) Sensor
B) Transmitter
C) Indicator
D) Transformer
3. Which of the following is a static characteristic of an instrument?
A) Response time
B) Hysteresis
C) Sensitivity
D) Frequency response
4. The ability of an instrument to produce the same output for the same input repeatedly is called:
A) Accuracy
B) Repeatability
C) Precision
D) Resolution
5. Which of the following is a bioelectrical signal?
A) Blood pressure
B) EEG
C) Body temperature
D) Respiration rate
6. Which method for measuring blood pressure is indirect and commonly used clinically?
A) Auscultatory method
B) Catheter insertion
C) Oscillometric method
D) Electronic manometer
7. Which method uses injection of an indicator and measurement of its dilution to estimate cardiac output?
A) Thermal dilution
B) Dye dilution
C) Indicator dilution
D) All of the above
8. Plethysmography is used to measure:
A) Lung volumes including residual volume
B) Tidal Volume
C) Forced expiratory volume
D) Functional residual capacity
9. Bioelectric signals are generated due to:
A) Mechanical forces
B) Electrical activity of cells
C) Magnetic fields
D) Chemical pressure
10. Which ion primarily causes depolarization during an action potential?
A) Potassium (K^+)
B) Calcium (Ca^{2+})
C) Sodium (Na^+)
D) Chloride (Cl^-)
11. Needle electrodes are typically used for:
A) ECG recording
B) EMG recording from muscles
C) Blood pressure measurement
D) Measuring skin temperature
12. Electrode lead wires can pick up noise mainly due to:
A) Magnetic interference
B) Static electricity
C) Movement artifacts
D) All of the above

13. Einthoven's triangle is formed by:
 A) Three limb leads (RA, LA, LL)
 B) Chest leads only
 C) Precordial leads
 D) Frontal and transverse planes
14. The QRS complex in ECG corresponds to:
 A) Atrial depolarization
 B) Ventricular depolarization
 C) Ventricular repolarization
 D) Atrial repolarization
15. How many electrodes are typically used in the standard 10-20 EEG system?
 A) 21
 B) 10
 C) 30
 D) 20
16. Alpha waves are most prominent during:
 A) Deep sleep
 B) Mental concentration
 C) Awake, relaxed state with closed eyes
 D) REM sleep
17. Which component in an MRI scanner generates the strong magnetic field?
 A) Gradient coils
 B) Radiofrequency coil
 C) Superconducting magnet
 D) X-ray tube
18. The main detector in a CT scanner detects:
 A) Transmitted X-rays after passing through the body
 B) Electrical signals from magnetic resonance
 C) Heat emitted from tissues
 D) Reflected sound waves
19. Which part of a pacemaker generates electrical pulses?
 A) Pulse generator
 B) Leads
 C) Battery
 D) Capacitor
20. Endoscopes typically have:
 A) Magnetic coils
 B) Fiber optic light and camera system
 C) Ultrasound crystals
 D) X-ray tubes

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Define: (i) Resolution (ii) Accuracy (iii) Drift (iv) Hysteresis .	8	L*2	1	1
b) A patient is suspected of having high blood pressure that fluctuates throughout the day. Which method would you recommend for continuous monitoring and explain the measurement method.	8	L3	2	5
2. a) Explain the principle, construction, and working of a resistive strain gauge. Explain its application in biomedical instrumentation.	8	L2	1	1
b) A hospital is considering installing electromagnetic flow meters in its intensive care units. Discuss how these meters function and evaluate their usefulness and limitations in real-time blood flow monitoring.	8	L3	2	5
3. a) Explain the various functional elements of an instrumentation system.	8	L2	1	1
b) Describe how blood pH, pO ₂ , and pCO ₂ are measured using a blood gas analyzer.	8	L2	2	5

Unit – II

4. a) Explain the ionic basis of resting membrane potential in neurons. How is it maintained?	8	L2	3	1
b) Explain Einthoven's Triangle and its significance in electrocardiography (ECG). How does it help in recording the standard limb leads?	8	L2	4	5
5. a) Discuss the advantages and limitations of using pregelled disposable electrodes compared to reusable electrodes. In what scenarios are they most effective?	8	L2	3	1
b) Differentiate between unipolar, bipolar, and average reference modes in EEG recordings. What are the advantages and limitations of each?	8	L2	4	5

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|-------|---|---|----|---|---|
| 6. a) | Explain how a differential amplifier can be used to reduce common-mode noise. | 8 | L2 | 3 | 1 |
| b) | What is electroretinography (ERG)? Describe how unipolar and bipolar modes are used in ERG recordings and their relevance in ophthalmology. | 8 | L2 | 4 | 5 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the basic principle and functional components of an MRI (Magnetic Resonance Imaging) system. How does it produce detailed images of soft tissues? | 8 | L2 | 5 | 5 |
| b) | Explain the working mechanism and functional parts of a defibrillator. How does it restore normal heart rhythm during cardiac arrest? | 8 | L2 | 5 | 5 |
| 8. a) | Explain the concept of thermography in medical imaging. What are its functional components, and in which clinical scenarios is it commonly used? | 8 | L2 | 5 | 5 |
| b) | Discuss the principle and components of a medical ventilator. How is it used to support patients with respiratory failure? | 8 | L2 | 5 | 5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
